

Project Planning Phase

Project Planning Template

(Product Backlog, Sprint Planning, User Stories, Story Points)

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Milestones:

Milestone	Description	Target Date	Status
M1	Data Collection and Model Selection	28-29 June 2026	Completed
M2	Data Preprocessing and Model Training	30 June 2026	Completed
M3	Model Saving and Core Logic (app.py)	1 July 2026	Completed
M4	Frontend Development (Streamlit App)	2 July 2026	Completed
M5	Deployment and Testing	3 July 2026	Completed
M6	Documentation and Submission	4 July 2026	Completed

Sprint Delivery Plan (Product Backlog):

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member

Sprint-1	Data Collection	USN-1	As a developer, I collect and load the Crop Recommendation Dataset (2200 records, 7 features, 22 crop labels)	2	High	Duddu Vijaya Padma Priya
Sprint-1	Model Training	USN-2	As a developer, I train a Random Forest Classifier on 80% training data and evaluate accuracy on 20% test data	3	High	Duddu Vijaya Padma Priya
Sprint-2	Model Saving	USN-3	As a developer, I save the trained model as model.pkl using pickle for reuse without retraining	1	High	Kasa Sahitya

Sprint-2	Frontend Development	USN-4	As a developer, I build a Streamlit input form with 7 numeric fields and a Recommended Crop button	2	High	Kasa Sahitya
Sprint-2	Prediction Logic	USN-5	As a developer, I connect the form to model.pkl and display the predicted crop with confidence score and top-3 alternatives	2	High	Kasa Sahitya
Sprint-3	Deployment	USN-6	As a developer, I deploy the app on Streamlit Community Cloud and obtain a public URL	2	Medium	Duddu Vijaya Padma Priya
Sprint-3	Testing	USN-7	As a developer, I perform functional and performance testing and document results	2	Medium	Kasa Sahitya

Sprint-3	Documentation	USN-8	As a developer, I complete all 8 GitHub folders and submit the project	3	High	Duddu Vijaya Padma Priya
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Project Progress Tracker:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Actual Release Date
Sprint-1	5	2 Days	28 June 2026	29 June 2026	5	29 June 2026
Sprint-2	5	2 Days	30 June 2026	1 July 2026	5	1 July 2026
Sprint-3	7	3 Days	2 July 2026	4 July 2026	7	4 July 2026

Team Management:

Team Member	Role	Responsibilities	Status
Duddu Vijaya Padma Priya	Active Member	Data collection, model training, deployment, documentation Phases 1-4 & 7	Active
Kasa Sahitya	Active Member	Frontend development (app.py), testing, GitHub upload, documentation Phases 5-6 & 8	Active

Note: The project was completed by the 2 active members (Duddu Vijaya Padma Priya and Kasa Sahitya). The originally assigned team lead and other team members were inactive throughout the program.